AL RYADA UNIVERSITY FOR SCIENCE AND TECHNOLOGY

FACULTY OF NURSING

CRITICAL AND EMERGENCY NURSING First Term

Medical Surgical Nursing Department 2025-2026 Practical Guide

University Vision

Al-Ryada University for Science and Technology aspires to be a distinguished and highly competitive leading university at the local and international levels in line with the sustainable development goals.

University Mission

The University is committed to providing a distinguished graduate capable of innovating and competing in the labour market and meeting the challenges of the future by providing a stimulating educational environment with intelligent and sophisticated techniques and strengthening partnerships with local and international universities and institutions and employing scientific research to achieve academic and research excellence while adhering to national identity, ethical values and community responsibility.

Vision of the College:

The College of Nursing at the Al-Ryada University for Science and Technology in Egypt aspires to become a leading college in embracing comprehensive, distinguished education that supports students' personal capabilities and leads to self-confidence and continuous self-development. This positively impacts fair professional competition locally, regionally, and internationally, and advances the educational, research, and societal aspects of the profession.

Mission of the College:

The College of Nursing at the Al-Ryada University for Science and Technology in Egypt is committed to utilizing a variety of modern educational methods and theories and continually developing them in line with scientific developments and educational quality standards. This will result in the graduation of distinguished cadres with a high competitiveness at the local, regional, and international levels in the practice of the nursing profession and in the field of scientific research. The College is committed to actively participating in the continuous development of various aspects of the profession, which will positively impact the profession and society, achieving sustainable development.

Course Specification

1- Basic Information

Course Title (according to the bylaw)	**Clinical Critical Care and Emergency Nursing**			
Course Code (according to the bylaw)	**CLN 2422**			
Department/s participating in delivery of the course	**Critical Care and Emergency Nursing department**			
Number of credit hours/points of the course (according to the bylaw)	**Theoretical**	**Practical**	**Other (specify)**	**Total**
		3		
Course Type				
Academic level at which the course is taught	الفرقة/المستوي الثاني			
Academic Program	**Bachelor of Science in Nursing**			
Faculty/Institute	**Nursing**			
University/Academy	**AL Ryda for science and technology**			
Name of Course Coordinator	**Assistant professor /Afaf Mohammed Mansour**			
Course Specification Approval Date	**8/10/2025**			
Course Specification Approval (Attach the decision/minutes of the department /committee/council ....)				

1. Course Overview (Brief summary of scientific content)

This course focuses on the integration of knowledge and skills to provide safe and effective nursing care to critically ill adults patients with complex needs. Also, emergency nursing care is designed to provide nursing students with the triage skills required to care effectively and safely for seriously injured victims.

2. Course Learning Outcomes CLOs

3- Course specification based on competency:

Domain No. (1) PROFESSIONAL AND ETHICAL PRACTICE

Competency	Key elements	Course Subjects	Course Objectives
1.1.***Demonstrate knowledge, understanding, responsibility and accountability of the legal obligations for ethical nursing practice***	1.1.1Demonstrate understanding of the legislative framework and the role of the nurse and its regulatory functions. 1.1.2 Apply value statements in nurses' code of ethics and professional conduct for ethical decision making. 1.1.3 Practice nursing in accordance with institutional/national legislations, policies and procedural guidelines considering patient/ client rights. 1.1.4 Demonstrate responsibility and accountability for care within the scope of professional and practical level of competence	• Oropharyngeal airway insertion • Central Venous Pressure (CVP)	• Apply Oropharyngeal airway insertion • Perform measure Central Venous Pressure (CVP)

Domain No. (2) HOLISTIC PATIENT-CENTERED CARE

Competency	Key elements	Course Subject	Course Objectives
2.1**Provide holistic and evidence-based nursing care in different practice settings**.	2.1.1 Conduct holistic and focused bio-psychosocial and environmental assessment of health and illness in diverse settings	Basic life support (BLS)	- Apply ethical principles during CPR &code management -Apply Cardioversion for patients with Defibrillation - Management of patients in Cardiopulmonary Arrest - Make clinical decisions during cardiac arrest to save patient life
	2.1.2.Provide holistic nursing care that addresses the needs of individuals, families and communities across the life span	Arterial Blood Gases (ABG) and Interpretation	-Perform Arterial Blood Gases (ABG) -Interpret Arterial Blood Gases (ABG -Perform Special consideration when obtain Arterial Blood Gases (ABG

	2.1.6. Examine evidence that underlie clinical nursing practice to offer new insights to nursing care for patients, families, and communities	- Endotracheal tube insertion and care	- Apply Endotracheal tube insertion -Perform nursing care patients with endotracheal tube
2.2 Provide health education based on the needs/problems of the patient/client within a nursing framework	2.2.1.Determine health related learning needs of patient/client within the context of culture, values and norms.	- Suctioning	- Implementation suction patients with airway obstruction of secretion
	2.2.2.Assess factors that influence the patient's and family's ability, including readiness to learn, preferences for learning style, and levels of health literacy	Chest Tube	- Apply nursing intervention for critically ill patients with respiratory injury

3. Teaching and Learning Methods

1. **Demonstration**
2. **Redemonstration**..
3. **Case Scenario**
4. **Simulation**
5. **Audiovisual Material (Data show /education video)**
6. **Online Methods for teaching as (whatsapp, Telegram**

Course Schedule:

Number of the Week	Scientific content of the course (Course Topics)	Total Weekly Hours	Expected number of the Learning Hours			
			Theoretical teaching (lectures/discussion groups/ ......)	Training (Practical/Clinical/ ......)	Self-learning (Tasks/ Assignments/ Projects/ ...)	Other (to be determined)
1	Arterial puncture and Arterial blood gases with interpretation	6		1		
2	Central venous pressure	6		1		
3	Basic Life Support	6		1		
4	Defibrillation & Cardioversion	6		1		
5	Oropharyngeal Airway insertion and care	6		1		
6	Endotracheal intubation	6		1		
7	Endotracheal Tube Care	6		1		
8	Suctioning	6		1		
9	Chest tube drainage system	6		1		

• Learning Resources and Supportive Facilities *

Learning resources (books, scientific references, etc.) *	**The main (essential) reference for the course** (must be written in full according to the scientific documentation method)	• **Urden ,L., Stacy, K.,& Lough, M.(2020).** Priorities in Critical Care Nursing. 8 th.ed. ELSEVIER Company • **Gohnson ,A.,& Crumiett ,H.**,(2018). Critical Care Nursing Certification . 7th .ed. New York • **Good,V., and Kirkwood P.(2018).** Advanced Critical Care Nursing. Elsevier company. 2nd Editition ; 464: 67. • **Wiegand ,D., (2017).** Procedure Manual For High Acuity, Progressive ,and Critical Care .7th .ed. Elsevier company.
	Other References	
	Electronic Sources (Links must be added)	• **Nursing Informatics in Critical Care**A systematic review published in BMC Nursing explores how nursing informatics—like electronic health records (EHRs), clinical decision support systems (CDSSs), and telehealth—

		• **The** American Association of Critical-Care Nurses (AACN) **offers several journals and newsletters:**
	Learning Platforms (Links must be added)	**1. CCRN (Critical Care Registered Nurse) – AACN** **2. CEN (Certified Emergency Nurse) – BCE** **3. TNCC (Trauma Nursing Core Course) – ENA** **4. ACLS & PALS (Advanced Cardiac/Pediatric Life Support) – AHA** **5. CCNS (Critical Care Nurse Specialist) – AACN**
	Other (to be mentioned)	-
Supportive facilities & equipment for teaching and learning *	**Devices/Instruments**	**Montor, ECG, DC Shock, ambo bag**
	Supplies	**Syring, cvp manometer**
	Electronic Programs	**LMS, Microsoft team, Zoom meeting**
	Skill Labs/ Simulators	**Sem man**
	Virtual Labs	**Glasses for virtual reality**
	Other (to be mentioned)	-

7. Methods of students' assessment

No.	Assessment Methods *	Assessment Timing (Week Number)	Marks/ Scores	Percentage of total course Marks
1	Exam 1written (quiz 1)	4weeks	7.5	
2	Exam 2 ……. (quiz 2)	8weeks	7.5	
3	Midterm exam	7week	10	
4	Final Practical/Clinical/… Exam	15 weeks	60	
5	Final Oral Exam	15week	20	
6	Assignments / case study	10 weeks	10	
7	Field training (Assessment-care plan-punctuality)		30	
8	Activity	-	5	
	Total		150	

* The list mentioned is an example, the institution may add and/or delete depending on the nature of the course

Name and Signature Course Coordinator	Name and Signature Program Coordinator

Table of content

Arterial Puncture (ABG Sampling)

Outlines: -

- **Definition of Arterial blood sampling**
- **Purposes of Arterial blood sampling**
- **Indications of Arterial blood sampling**
- **Contra indications of Arterial blood sampling**
- **Equipment for Arterial blood sampling**
- **Procedure of Arterial blood sampling**

Definition

Arterial puncture is a procedure to obtain arterial blood, usually from the radial, brachial, or femoral artery, for **arterial blood gas (ABG) analysis** to assess oxygenation, ventilation, and acid-base balance.

Purpose

- To evaluate:
  - **Oxygenation status (PaO_2)**
  - **Ventilation status ($PaCO_2$)**
  - **Acid-base balance (pH, HCO_3^-)**
  - Response to oxygen therapy or ventilator support
- To monitoring critically ill patients with respiratory or metabolic disorders.

Indication.

1. Respiratory Acidosis
2. COPD exacerbation
3. Respiratory Alkalosis
4. Anxiety-induced hyperventilation
5. Diabetic ketoacidosis

Contraindications

- Negative Allen's test (poor collateral circulation)
- Local infection or skin lesions at puncture site
- Severe peripheral vascular disease
- Coagulopathy or anticoagulant therapy (relative contraindication)

Equipment Needed

- Sterile gloves
- ABG kit (heparinized syringe with needle)
- Alcohol swabs or antiseptic solution
- Sterile gauze pads
- Tape or adhesive bandage
- Ice or transport medium for sample
- Local anesthetic (optional)
- Sharps disposal container

Patient Preparation

- Explain the procedure to reduce anxiety.
- Check **physician's order** and verify patient identity.
- Assess for contraindications.
- Perform **Allen's test** for radial artery site to check collateral circulation.
- Position patient comfortably with the selected arm supported and wrist hyperextended (using a rolled towel under wrist).
- Provide **oxygen if indicated** and ensure patient is stable.

Nursing Considerations

- Monitor for complications:
  - Hematoma
  - Arterial spasm
  - Thrombosis
  - Infection

- Notify provider if persistent bleeding or circulation impairment occurs.
- Ensure patient comfort and reassurance throughout.

Potential Complications

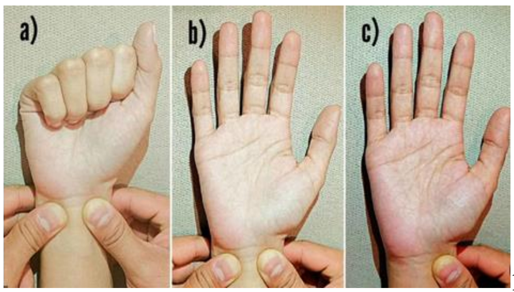
- Hematoma
- Arterial occlusion
- Infection
- Air embolism (if air bubbles not removed)

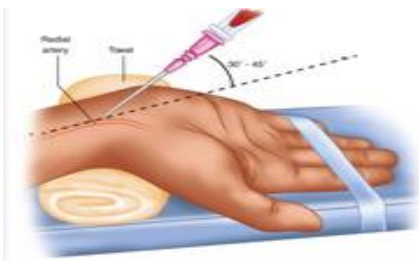
Special consideration (with Rationale)

- **Choose alternate site** (other radial, brachial, femoral) if Allen's test is negative or radial pulse is weak.
  Rationale: Reduces ischemic risk and improves sampling success.
- **Avoid repeated attempts in same spot; limit total attempts** (e.g., ≤2 per clinician, then escalate).
  Rationale: Minimizes hematoma, pain, and arterial damage.
- **If severe spasm occurs**, warm the area, gently massage proximal to site, and pause before reattempting elsewhere.
  Rationale: Heat and time relieve vasospasm, improving safety.
- **If patient on high-flow O_2 or ventilator**, record **FiO_2 and ventilation mode/settings** and ensure the patient has been on the **current settings ≥10 minutes** before sampling (unless urgent).
  Rationale: Stabilizes alveolar gas tensions for a representative sample.
- Use smallest feasible needle **per kit/protocol (commonly 23–25G for radial).**
  ***Rationale:* Reduces pain and vascular trauma while allowing adequate flow**

Arterial Puncture (ABG sampling) procedure

Step	Rationale
Verify order & indications (ABG reason, FiO_2/vent settings).	➢ Ensure the test is clinically indicated and interpreted in the correct oxygenation/ventilation context.
Identify patient using 2 identifiers.	➢ Prevents mislabeling and serious diagnostic/therapeutic errors.
Review contraindications/risks (negative Allen's test, infection at site, AV fistula, severe PAD, coagulopathy/anticoagulants, thrombocytopenia).	➢ Minimizes harm by choosing safest site and anticipating bleeding or ischemic complications.
Check recent medications (anticoagulants, antiplatelet) & **platelets/INR** if available.	➢ Informs hemostasis strategy (longer pressure time, alternate site).
Explain the procedure (brief pain, pressure after, keep still). Obtain consent per policy.	➢ Reduces anxiety, improves cooperation, and meets legal/ethical standards.
Hand hygiene & appropriate PPE.	➢ Reduces infection transmission risk.
Gather equipment: heparinized ABG syringe/needle (or pre-heparinized kit), alcohol/chlorhexidine swabs, sterile gauze, tape/occlusive dressing, sharps container, labels/ice (if lab requires), local anesthetic	➢ Ensures uninterrupted, efficient, and safe procedure; prevents sample clotting and contamination.

(1% lidocaine without epinephrine) optional per policy, rolled towel for wrist support.	
Position patient supine or semi-Fowler; **extend wrist** ~30° with a rolled towel under it; palm up.	➢ Brings radial artery superficial/taut, improving success and reducing multiple attempts.
Perform Allen's test (for radial site): occlude radial & ulnar, have patient clench/unclench,  release ulnar—hand should flush within ~5–10 s.	➢ Confirms adequate **collateral circulation** via ulnar artery, reducing risk of hand ischemia if radial flow is compromised.
Prepare the syringe: expel excess heparin (if not dry-heparinized), ensure needle secure, remove air.	➢ Prevents dilution artifacts (excess liquid heparin) and air affecting gas values.
Skin antisepsis: vigorous friction with alcohol or chlorhexidine; allow to dry completely.	➢ Lowers skin bioburden and prevents alcohol-induced hemolysis/electrical interference; drying time maximizes antiseptic effect.
Optional local anesthesia: small intradermal bleb of 1% lidocaine without epinephrine; aspirate before injecting.	➢ Significantly reduces pain without altering ABG gas tensions when used properly.

B. During the Procedure (Sampling)	
Palpate and locate the point of maximal radial pulse (non-dominant hand preferred if adequate).	➢ Maximizes first-pass success; reduces trauma and spasm.
Stabilize the artery with non-dominant hand; **insert needle bevel up** at **30–45°** toward pulse. 	➢ Optimal angle enters the lumen with minimal wall trauma; bevel up facilitates arterial flow into syringe.
Do NOT aspirate; allow **arterial pressure to fill** the syringe (1–2 mL typically adequate).	➢ Avoids hemolysis and negative-pressure artifact; confirms arterial placement (pulsatile fill).
If no flash/flow, withdraw slightly or change angle minimally while maintaining skin entry site; do not "fish."	➢ Small adjustments can regain lumen without creating multiple punctures or hematoma.
Once volume adequate, withdraw needle smoothly and **immediately apply firm pressure** with sterile gauze for **≥5 minutes** (≥10–15 min if anticoagulated or coagulopathy).	➢ Prevents bleeding and hematoma; arterial pressure is higher than venous, so longer compression is required.
Expel air bubbles from syringe; **cap/secure** needle or use safety device.	➢ Air bubbles **alter PaO_2/$PaCO_2$** via gas exchange, leading to inaccurate results; sharps safety prevents injury.

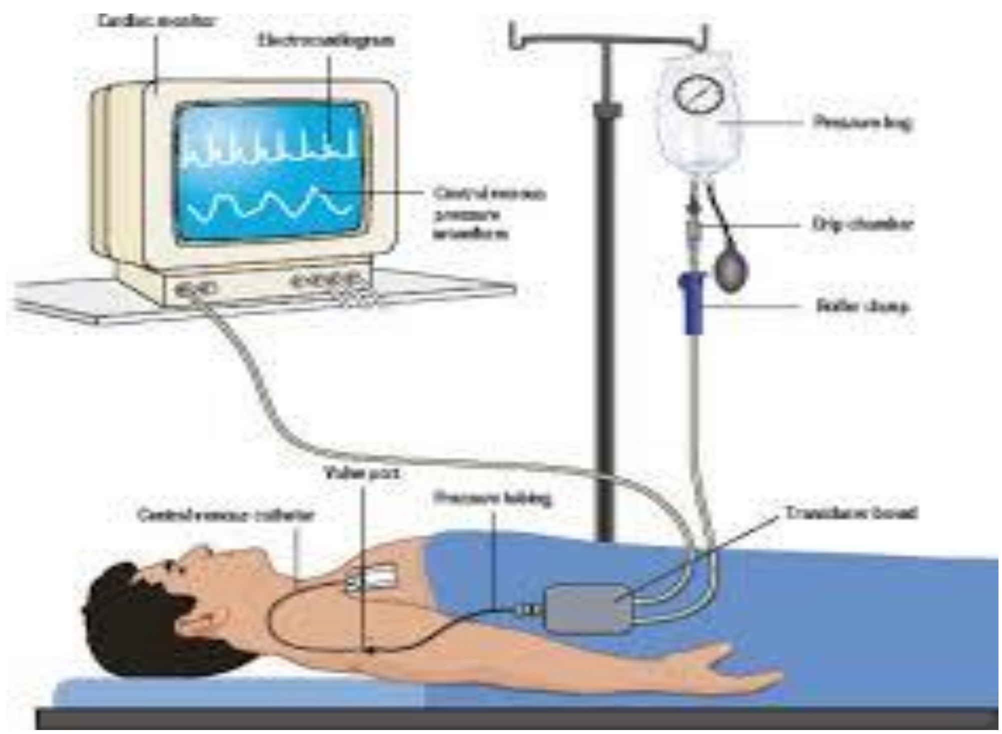
Gently roll/invert syringe to mix blood with heparin (if liquid heparin used).	➢ Prevents micro clot formation that can occlude analyzers and invalidate results.
Assess patient during/after puncture for pain, pallor, numbness, paresthesia, or color change distally.	➢ Early detection of vasospasm, nerve irritation, or compromised perfusion allows timely intervention.
B. C. After the Procedure (Label, Transport, Monitor)	**Rational**
Label the sample at bedside with 2 identifiers, date/time, site, **FiO_2/vent settings**, and patient position.	➢ Ensures traceability and correct interpretation (gases depend on oxygen delivery and posture).
Place on ice or **send immediately** according to lab policy and distance to analyzer.	➢ Slows cellular metabolism (O_2 consumption/CO_2 production) and preserves accuracy when immediate analysis isn't possible.
Check the puncture site before releasing pressure; **apply occlusive dressing**.	➢ Confirms hemostasis and protects site from contamination/bleeding.
Reassess distal perfusion (capillary refill, warmth, color, pulse) and **document**.	➢ Detects delayed ischemia or hematoma formation; documents neurovascular status.
Dispose of sharps and waste properly; hand hygiene.	➢ Prevents needle stick injuries and infection spread.
Document: site, side, number of attempts, patient tolerance, complications, post-procedure	➢ Provides a complete legal/clinical record and supports interpretation of results.

assessment, oxygen/vent settings, and whether ice was used.	
Monitor the patient for ongoing bleeding, hematoma, numbness, increasing pain, or color change for at least 10–15 minutes.	➢ Ensures early recognition and management of complications (e.g., compress, elevate, notify provider).

Central Venous Pressure (CVP)

Outlines: -

- **Definition of central venous pressure**
- **Indications of central venous pressure measurement**
- **Complications of central venous pressure**
- **Equipment for central venous pressure**
- **Procedure of central venous pressure measurement**



Definition

"CVP is the pressure of blood in the central veins, measured at the level of the right atrium.

Purpose

- **To assess the circulating blood volume and the heart's ability to pump blood back into circulation."**

Indications of cvp

Indication	Description
Critically ill / ICU monitoring	Assesses overall hemodynamic status and guides sepsis resuscitation
Fluid resuscitation / major surgery	Guides volume management in shock or significant fluid shifts
Cardiac dysfunction evaluation	Helps detect tamponade, right heart failure, or infarction
Administration of high-risk IV therapies	For vasoactive drugs, TPN, hyperosmolar or irritant solutions
Postoperative monitoring	Tracks stability after significant surgical interventions
When peripheral access is inadequate	Provides essential central venous access in emergency settings

Methods of CVP Measurement:

Measurement Method	Description	Key Considerations
Water Manometer (Invasive)	Fluid column reflects CVP in cm H_2O	Simple, but manual and sensitive to positioning
Electronic Transducer (Invasive)	Real-time waveform and digital measurement (mmHg)	Requires calibration and correct leveling
JVP Observation (Non-invasive)	Visual estimation based on neck vein height	Low precision, operator-dependent
Ultrasound IVC/IJV (Non-invasive)	Measures diameter and collapsibility as CVP surrogate	More accurate than visual, still indirect
Plethysmography (Forearm)	Forearm volume changes used as proxy for CVP	Experimental, high correlation reported

Complication of CVP insertion: -

Mechanical - Arterial puncture, pneumothorax, cardiac perforation/tamponade, air embolism,

- malposition, nerve injury

Infectious: - Catheter-related bloodstream infections (sepsis)

- Catheter-related thrombosis,

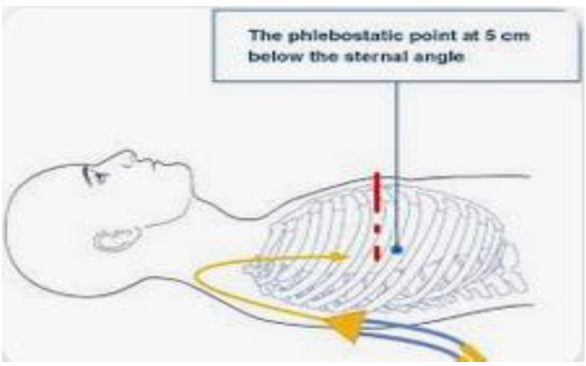
Thrombotic : pulmonary embolism

Equipment:

1. CVL
2. IV fluid as normal saline
3. IV pole
4. 3 ways
5. Gloves

Central Venous Pressure (CVP) Monitoring –Procedure

Step	**Rationale**
A. Preparation Phase	
1.Verify the physician's order	❖ Legal and clinical accountability
2.Identify patient using two identifiers	❖ Patient safety
3.Explain the procedure to the patient	❖ Reduces anxiety and ensures cooperation
4.Perform hand hygiene and don gloves/PPE	❖ Prevents healthcare-associated infection
5.Gather and inspect all equipment	❖ Ensure everything is sterile, functional, and within expiration dates
6.Assess **insertion site** of central line for signs of infection or dislodgement	❖ Prevent further complications or false readings

B. Patient Positioning and System Setup	
7.Position patient **supine** or **semi-Fowler's** (0–45°)	❖ Position significantly affects pressure readings
8.Identify **phlebostatic axis** (4th intercostal space, mid-axillary line)	❖ Standard reference for zeroing pressure systems
9.Secure transducer or manometer at the  level of the phlebostatic axis	❖ Ensures accuracy in relation to the right atrium
10.Zero the transducer (if using electronic system) □ use **a local/manual system**, you typically measure CVP using a **manometer** instead of an electronic transducer. □ In this case, there is **NO zeroing of a transducer**, because the manometer works based on **gravity and fluid column pressure**, not electronic sensors.	❖ Removes atmospheric pressure from the equation

☐ Instead, you: • Position the **zero-reference point (phlebostatic axis)** correctly at the mid-axillary line, **4th intercostal** space. • Connect the manometer to the CVP line via a three-way stopcock. • Fill the manometer with fluid to the zero mark, then allow it to equilibrate with the patient's venous pressure. • Read the CVP where the fluid level stabilizes at the end of expiration.	
11.Prime IV tubing and flush system with saline	❖ Air can alter pressure readings or lead to embolism
C. Measuring CVP	**Rationale**
12.Connect CVP line to **manometer** or transducer system via stopcock 	❖ Prevents contamination
13.For manometer: Fill column to 20–25 cm H_2O using IV bag	❖ Prevent fluid overload or backflow

14.Turn stopcock to open connection between manometer and patient	❖ CVP = height where fluid column stabilizes
15.Read and record the CVP level at end-expiration	❖ Most accurate point due to minimal intrathoracic pressure changes
16.Return stopcock to IV fluids or flush line	❖ Maintains line patency and patient safety
17.Compare readings with baseline and trend over time	❖ Single readings are not as valuable as trends

D. Post-Procedure Care	**Rationale**
18.Assess for complications: infection, bleeding, air embolism, arrhythmia	❖ early detection prevents serious harm
19.Change dressings or secure connections if needed	❖ Maintain sterile barrier
20.Dispose of used supplies in appropriate containers	❖ Prevents cross-contamination
21.Remove gloves, perform hand hygiene	❖ Break the transmission chain
22.Document: CVP reading, patient position, respiratory phase, line status, and any complications	❖ Provides legal, accurate record and communication for care team

Interpretation of CVP Values

CVP (cm H_2O)	Interpretation
2–6 cm H_2O	❖ Normal range
<2 cm H_2O	❖ Hypovolemia, vasodilation
>6 cm H_2O	❖ Hypervolemia, right heart failure, tamponade, positive pressure ventilation

Interpretation of CVP Values

CVP Value	What It Means
0–6 mmHg	**Normal** CVP (normal blood volume & pressure)
< 0 mmHg	**Low** CVP → May mean dehydration or blood loss
> 8 mmHg	**High** CVP → May mean fluid overload, heart failure, or lung pressure

Basic life support (BLS)

Outlines: -

- **Definition**
- **Chain of survival**
- **BLS Algorithm**

Definition:

Basic Life Support (BLS) refers to the care healthcare providers and public safety professionals provide to patients who are experiencing **respiratory arrest**, **cardiac arrest**, or **airway obstruction**.

BLS includes **psychomotor skills** for:

- Performing high-quality **cardiopulmonary resuscitation (CPR)**
- Using an **automated external defibrillator (AED)**
- Relieving an **obstructed airway** for patients of all ages

BLS also focuses on the integration of the following key skills to help rescuers achieve optimal patient outcomes:

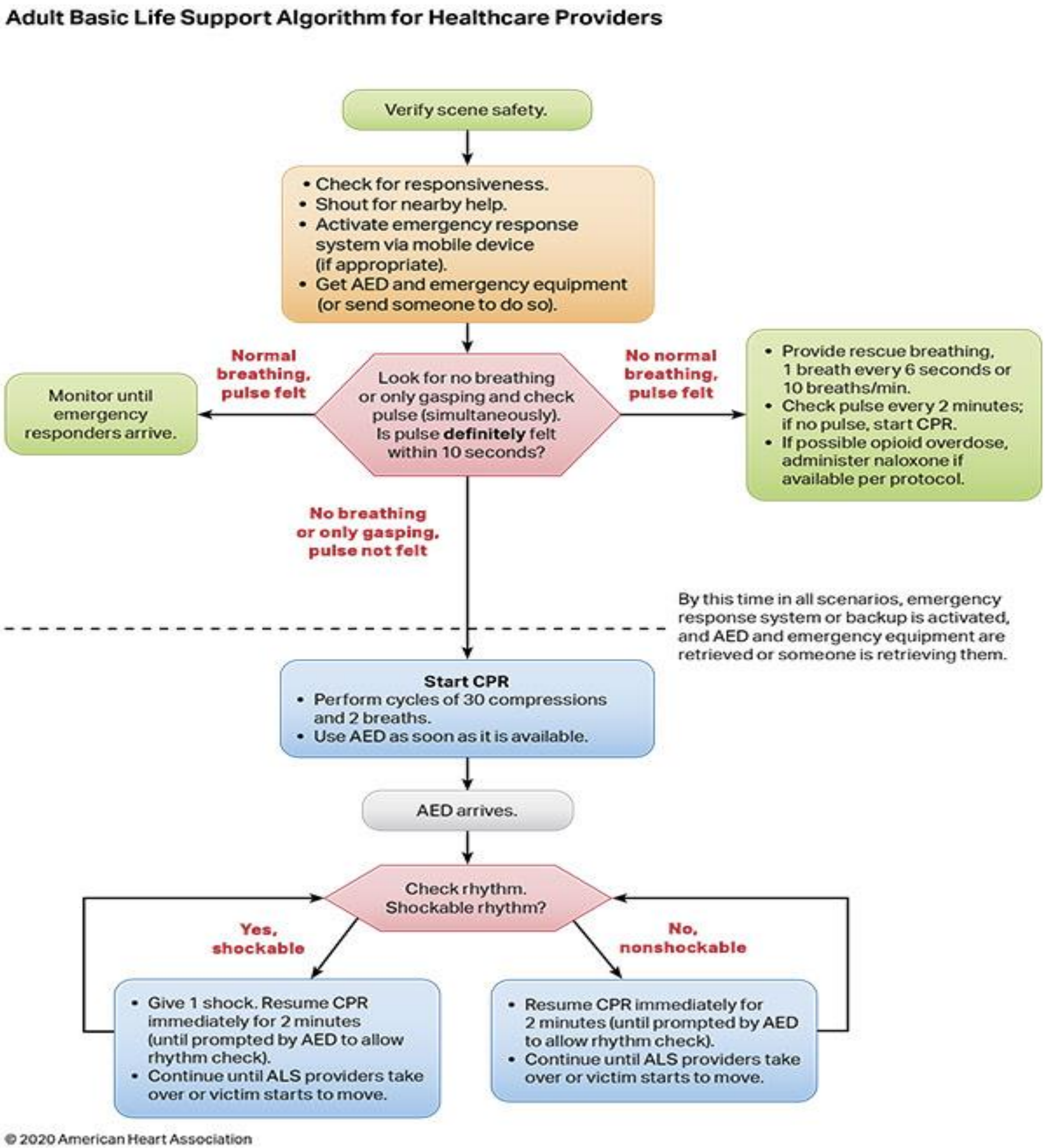
- **Critical Thinking**: Clear and rational thinking based on facts presented and the learner's experience and expertise
- **Problem Solving**: Identifying solutions to issues that arise using readily available resources
- **Communication**: A closed-loop process involving a sender, message, and receiver
- **Team Dynamics**: Integration and coordination of all team members working together toward a common goal

NB: The main aim is to provide oxygen to the brain to prevent irreversible cerebral damage. Cerebral damage starts after **4–6 minutes** without adequate perfusion.

Chain of Survival for Adult

- Immediate Recognition & EMS Activation
- Early CPR
- Rapid Defibrillation
- Effective Advanced Life Support
- Integrated Post Cardiac Arrest Care

BLS Algorithm

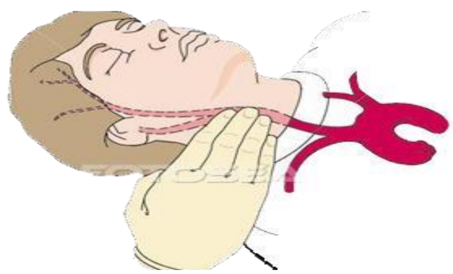


Verify Scene Safety, Check for Responsiveness, and Get Help

- ✓ Verify that the scene is safe for you and the victims; you do not want to become the second victim
- ✓ Check for responsiveness Tap the victim's shoulder and shout " you are OK"
- ✓ If the victim unresponsive shout for nearby help
- ✓ Activate emergency response system as appropriate in your setting Depending on your work situation call your local emergency number from your phone mobilize the code team or notify advanced life support
- ✓ If you are alone get the AED Defibrillator and emergency equipment If someone else available send that person to get it

2- Assess for Breathing and Pulse at the same time

- ➢ Check the carotid pulse for 5 to 10 sec
- ➢ From your side don't cross the neck
- ➢ Don't assess two side together



- ➢ Agonal gasping is not normal breathing
- ➢ Agonal Gasps may be present in the first minutes after sudden cardiac arrest
- ➢ The mouth may be open and the jaw head or neck may move with gasps

3 Chest compression

- ✓ Compress the center of the chest lower half of the sternum between the two nipples

- ✓ Compression ventilation ratio is 30 to 2
- ✓ Push Fast Push hard

NB Quality of Chest compression

- ❑ Allow complete chest compressions Avoid leaning on the chest between compressions to allow full chest wall recoil for adults in cardiac arrest
- ❑ Minimize interruptions not more than 10 sec
- ❑ Avoid excessive ventilation
- ❑ Rotate the compressor every 2 min
- ❑ Minimize interruptions not more than 10 sec Minimize the frequency and duration of interruptions in compressions to maximize the number of compressions delivered per min

4- Breathing

<u>A Mouth to Mouth Breath</u>

- Maintained open Airway
- Pinch the nose
- Seal the mouth with your mouth
- Look to victim chest rising

<u>B- Giving Adult Mouth to Mask Breath</u>

- To use a mask the lone rescuer is at the victim‘s side so you can give breaths and perform chest compression when positioned at the victims side
- The lone rescuer holds the mask against the patient face Good Sealing
- Open the airway with head tilt
- Chin lift and Give 2 Breathes

C- Using the bag-mask During 2 rescuer CPR

E-C clamp technique

5- Automated External Defibrillator AED

Power on the AED

Attach the AED pads

Clear the victim and analyze the rhythm

Make Sure you are not touching the patient while analyzing

The AED will advise you if the shock is advised

Defibrillation and Cardioversion

Outlines: -

- **Definition of cardioversion**
- **Definition of defibrillation**
- **Types of defibrillators**
- **Indications of D.C**
- **Equipment**
- **Patient assessment**
- **Patient preparation**
- **Procedure**

Definitions

Cardioversion: is a procedure consisting of 2 different applications, electrical or medical, performed to provide normal sinus rhythm in arrhythmic events. While medical cardioversion is preferred in stable patients and often in elective patients, electrical cardioversion is often preferred in patients who cannot get a medical response or who need an urgent CV

Defibrillation uses a therapeutic dose of electric current to the heart at a random moment in the cardiac cycle and is the most effective resuscitation measure for cardiac arrest associated with ventricular fibrillation and pulseless ventricular tachycardia.

Types of Defibrillator.

- **Monophasic:** Older technology; energy flows in one direction.
- **Biphasic:** Modern standard; energy flows in two directions, more effective for atrial and ventricular dysrhythmias.

Indications of Cardioversion

1- Atrial fibrillation- flutter
2- Supraventricular tachycardia due to reentry
3- Atrial tachycardia
4- Ventricular tachycardia with pulse

N.B Atrial fibrillation and flutter are the most common arrhythmias in which cardioversion is used.

Cardioversion Vs. Defibrillation

	Cardioversion	**Defibrillation**
Shock	**Synchronized shock** synced shock delivered only during the r wave of QRS	**Asynchronous** Done with an automated external defibrillation
Joules	**Lower** amount of joules (energy) used	**Higher** amount of joules (energy) used
CPR	Not done with CPR	Resume CPR after shock
Used for	Stable patients Must have QRS	Unstable patients
Examples	Atrial fibrillation Atrial flutter	VF – VT
Memory tricks	Cardioversion are **C**arefully planned	If it has a **V** (VT- VF) give them the **D** (defibrillation)

Equipment

- Defibrillator/
- monitor with electrocardiogram (ECG) /recorder capable of delivering a synchronized shock
- ECG cable
- Self-adhesive defibrillation pads connected directly to the defibrillator or conductive gel, paste, or prepackaged gelled conduction pads to be used with defibrillator paddles
- Intravenous sedative
- Bag-valve device with mask and oxygen delivery
- Flow meter for oxygen administration, oxygen source
- Emergency suction and intubation equipment
- Blood pressure monitoring equipment
- Pulse oximeter
- Intravenous infusion pumps
- Additional equipment, to have available as needed, includes the following:
- Cardiac board
- Emergency medications

Patient Assessment

1. Evaluate ECG for tachydysrhythmias.
2. Monitor vital signs and symptoms of hemodynamic instability.
3. Assess consciousness and peripheral pulses.
4. Check serum electrolytes, digitalis levels, and ABGs.

Patient Preparation

1. Confirm patient identity.
2. Educate patient and family; ensure informed consent (unless emergent).
3. Conduct time-out if not an emergency.
4. Obtain baseline 12-lead ECG.
5. Keep NPO as per policy.
6. Establish IV access.
7. Position patient supine.
8. Remove metallic objects and medication patches from chest.
9. Ensure dry environment and dry skin.
10. Remove chest hair if needed for pad contact.
11. Remove loose dentures if necessary.
12. Pre- oxygenate and sedate if time permits.
13. Maintain airway and oxygenation throughout procedure.

Defibrillation Procedure

	Steps	Rational
	Preparatory phase	
1)	Perform hand hygiene.	
2)	Prepare the patient and equipment (emergency cart with defibrillator on standby)	To ensure and quick flow of the procedure without interuptions; to save time and effort
	Performance phase	
3)	**A. Unmonitored patient** - check rhythm and pulse from an ECG or cardiac monitor to confirm abnormal rhythm	To assess if the rhythm is shockable
4)	Expose anterior chest and remove jewelry in the area	Jewelry may interfere with electrical current and cause severe burns
5)	Start CPR immediately while the 2nd persons gets defibrillator	TO Provide oxygenated blood supply to the cerebral and coronary arteries.
6)	Check if the defibrillator is on the unsynchronized mode.	Unsynchronized defibrillation is a high energy shock that is delivered as soon as the shock button is pushed on a defibrillator. The shock may fall randomly anywhere within the cardiac cycle (QRS complex). It is used where there is no coordinated intrinsic electrical activity in the heart (pulseless VT – VF) or the defibrillator fails to synchronize in an unstable patient.
7)	Apply interface material into the patient (gel pads) or to the paddle (gel paste), the electrode paddles should be in firm contact with the patient skin.	To prevent severe burns to the patient's chest. To prevent the diversion of current from travelling to the heart and to provide better conduction.
8)	Remove oxygen from immediate area	Prevents danger of fire explosions
9)	A second person should turn on the defibrillator to the prescribed setting.	Biphasic cardioversion has shown that less energy is required to convert an arrhythmia

	Usually 120-200 joules for biphasic defibrillator and 360 joules for monophasic defibrillator	to a normal sinus rhythm than monophasic cardioversion. Consequently use of biphasic energy results in fewer delivered shocks to the patient and less cumulative energy delivered. Potential benefits include fewer burn wounds, less tissue damage and reduced damage to heart muscle than is found in high voltage shocks.
10)	Apply paddles in two ways a. Anterolateral position: apply one electrode just to the right of the upper sternum below the clavicle and the other electrode just to the left to the cardiac apex or left nipple. b. anteroposterior position: the anterior paddle is held with pressure on the middle of the sternum while the patient lies on the posterior paddle under the left infrascapular region.	The paddles are placed so that the electrical current flows through as much of the myocardium as possible In this method , the current directly traverses the heart
11)	Grasp the paddles only with insulated handles. Avoid any body contact to the patient	To prevent getting shocked
12)	Charge the paddles, once the paddles are charged, give the command for personnel to stand clear of the patient and bed.	To prevent getting shocked
13)	If using handheld paddles, apply pressure to each paddle against the chest wall and depress both buttons on the paddles simultaneously and hold until the defibrillator fires.	Firm paddle pressure decreases transthoracic resistance, improving the flow of electrical current across the heart.

14)	Remove the paddles from the patient immediately after the shock is administered.	To avoid skin irritation and giving continuous shock to patient.
15)	Resume CPR until stable rhythm, spontaneous respiration, pulse return.	To oxygenate the patient and restore circulation.
16)	If the patient is still in ventricular Fibrillation or pulseless VT, immediately charge the paddles to 360 J (monophasic).	Immediate action increases the chance of successful subsequent depolarization of cardiac muscle.
17)	If the second attempt is unsuccessful, continue CPR and immediately charge the paddles to 360 J (monophasic).	
18)	**B. Monitored patient** Assess pulse and rhythm from the cardiac monitor.	To assess if the rhythm is shockable.
19)	If pulseless VT – VF, start CPR at once while other person is preparing the defibrillator.	CPR is essential before and after defibrillation to ensure blood supply to the coronary and cerebral arteries.
20)	Defibrillate patient as indicated.	To get out of a potentially fatal abnormal heart rhythm
21)	Analyze rhythm after 5 cycles of CPR or 2 minutes.	To determine if pulseless VT- VF is terminated and if pulse has been restored.
22)	**Follow up phase** If self-adhesive defibrillation pads were used, evaluate the placement and integrity of pads.	Self-adhesive defibrillation pads may crimp, crack, or fold with loss of adhesiveness.
23)	Discard used supplies in appropriate receptacle.	Reduces the transmission of microorganisms; standard precautions.
24)	Document all interventions done to the patient	To promote patient safety and ensure continuity of care

Procedure of cardioversion

No	Steps	Rational
1)	Perform hand hygiene.	Prevents transmission of microorganisms.
2)	Prepare the patient and equipment (emergency cart with defibrillator on standby)	To ensure and quick flow of the procedure without interuptions; to save time and effort.
3)	Connect the patient to the monitoring lead wires on the defibrillator. 	The R wave must be sensed by the defibrillator to achieve synchronization for cardioversion.
4)	Select a monitor lead that displays an R wave of sufficient amplitude to activate the synchronization mode of the defibrillator.	Synchronized cardioversion must sense the R wave to deliver the current outside the heart's vulnerable period. Lead II generally produces a large R wave.
5)	Place the defibrillator in synchronization mode. Ensure that the patient's QRS complexes appear with a marker to signify correct synchronization of the defibrillator with the patient's ECG rhythm.	Synchronization prevents the random delivery of an electrical charge, which may cause ventricular fibrillation. 
6)	If the defibrillator is unable to distinguish between the peak of the QRS complex and the peak of the T wave, as in polymorphic ventricular tachycardia, proceed with unsynchronized defibrillation.	Avoid a delay or failure of shock delivery in the synchronized mode. 

7)	Apply self-adhesive defibrillation pads or prepare defibrillation paddles. 	Reduces transthoracic resistance, enhancing electrical conduction.
8)	Pad or paddle placement in the patient with an ICD is the same as standard paddle placement for cardioversion. Pads or paddles should not be placed over the device.	Cardioversion over an implanted pacemaker or ICD may impair passage of current to the patient and may cause the device to malfunction or become damaged.
9)	Ensure that the defibrillator cables are positioned to allow for adequate access to the patient.	Allows cardioversion to occur without excessive tension on the cables.
10	Turn on the ECG recorder for a continuous printout.	Establishes a visual recording of the patient's current ECG status and response to intervention.
11	Charge the defibrillator as prescribed or in accordance with the recommendations of American Heart Association.	The defibrillator is charged with the lowest energy level necessary to convert the tachydysrhythmias.
12	Disconnect the oxygen source during actual cardioversion.	Decreases the risk of combustion in the presence of electrical current.
13	State "all clear" or similar wording three times and visually verify that everyone is clear of contact with the patient, bed, and equipment.	Maintains safety to caregivers because electrical current can be conducted from the patient to another individual if contact occurs.
14	Verify that the defibrillator is in the synchronization mode and that the patient's QRS complexes appear with a marker to signify synchronization of the defibrillator with the patient's ECG rhythm.	Synchronization prevents the random delivery of an electrical charge, which may potentiate correct ventricular fibrillation.
15	When using self-adhesive hands-free	Depolarizes the cardiac muscle; in

	defibrillation pads, depress the discharge button on the defibrillator to deliver the charge.	synchronized mode, a delay occurs to allow sensing of the QRS complex.
16	If using handheld paddles, apply pressure to each paddle against the chest wall and depress both buttons on the paddles simultaneously and hold until the defibrillator fires.	Firm paddle pressure decreases transthoracic resistance, thus improving the flow of electrical current across the axis of the heart.
17	Observe the monitor for conversion of the tachydysrhythmias and assess the patient's carotid pulse.	Simultaneous depolarization of the myocardial muscle cells should reestablish a single source of impulse generation.
18	Clean the defibrillator and remove any gel from the paddles.	Conductive gel accumulated on the defibrillator paddles impedes surface contact and increases transthoracic resistance.
19	If self-adhesive defibrillation pads were used, evaluate the placement and integrity of pads.	Self-adhesive defibrillation pads may crimp, crack, or fold with loss of adhesiveness.
20	Discard used supplies in appropriate receptacle.	Reduces the transmission of microorganisms; standard precautions.
21	Document all interventions done to the patient	To promote patient safety and ensure continuity of care.

Oropharyngeal airway insertion

Outlines: -

- **Definition of Oropharyngeal airway**
- **Purposes of Oropharyngeal airway insertion**
- **Contraindications of Oropharyngeal airway insertion**
- **Complications of Oropharyngeal airway insertion**
- **Equipment of Oropharyngeal airway insertion**
- **Procedure**

Definition

A curved plastic device is inserted into the mouth to the posterior pharynx to establish or maintain a patent airway. In an unconscious patient,

purpose of oropharyngeal airway: -:

To maintain airway patency for patients through the following instructions:

- An unconscious spontaneously breathing patient with an airway obstruction caused by an impaired gag reflex and a loss of tone to the submandibular muscles.
- Unsuccessful airway opening by other maneuvers, such as the head tilt, the chin lift, and the jaw thrust.
- A patient ventilated by a bag-valve-mask device. The oral airway elevates the soft tissues of the posterior pharynx, easing ventilation and minimizing gastric insufflations.
- An orally intubated patient bites/clenches the endotracheal tube; the oral airway is used as a bite block.
- An unconscious patient during suctioning, to facilitate the removal of a patient's oral secretions.

Contraindications

Absolute contraindications:

- Consciousness or presence of a gag reflex

Relative contraindications

Insertion of an oropharyngeal airway may not be feasible in some settings, such as:

- Oral trauma
- Trismus (restriction of mouth opening including spasm of muscles of mastication)

Nasopharyngeal airways may be used instead.

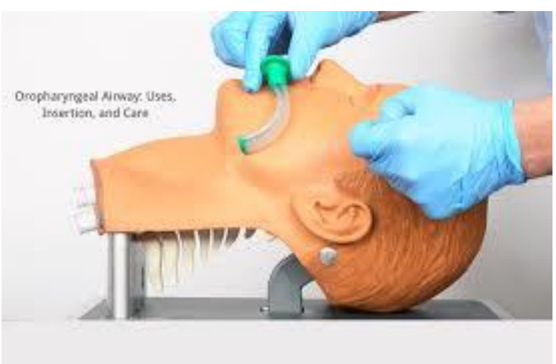
Complications:

1. Tooth damage or loss, tissue damage, and bleeding may result from insertion.
2. If the airway is too long, it may press the epiglottis against the entrance of the larynx, producing complete airway obstruction.
3. If the airway isn't inserted properly, it may push the tongue posteriorly, aggravating the problem of upper airway obstruction.

Equipment Needed: -

- Oropharyngeal suction equipment.
- Oropharyngeal airway.
- Tongue blade.
- Disposable gloves
- Mask

Procedure

Steps	Rational
1. Washing hands and applying gloves	Reduces the transmission of microorganisms.
2. Put the patient in the suitable position	Neck flexion with atlanto-occipital extension (sniffing position), or Neck in neutral position with spinal immobilization (cervical injury suspected)
3. Suction blood, secretions, or other foreign material from the patient's oropharynx.	The suction makes the airway clear and prevents aspiration.
4. Measure the distance from the central incisors to the angle of the mandible 	to approximate the correct size oral airway.

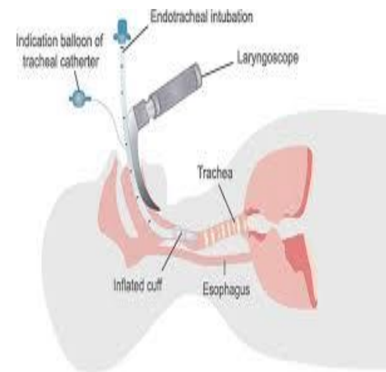
5. Select the appropriately sized oropharyngeal airway. 	if the oropharyngeal airway is too large, it may obstruct the larynx and if the oropharyngeal airway is too small or is inserted improperly, it pushes the tongue posteriorly, obstructing the airway.
6. Insert the airway with the curve pointing up into the oropharynx.	to avoid pushing the tongue toward the pharynx, and slide it over the tongue toward the back of the mouth
7. Use a tongue blade to depress and displace the tongue forward.	Use of the tongue blade prevents the airway from pushing the tongue backward during insertion.
8. Insert the airway upside down (with the curve pointing toward the back of the patient's head) into the mouth	to avoid pushing the tongue toward the pharynx, and slide it over the tongue toward the back of the mouth
9. The tip of the airway reaches the posterior wall of the pharynx, rotate the airway 180 degrees to the proper position	This technique prevents the airway from pushing the tongue backward during insertion and further obstructing the airway
10. Tape the air way in the position.	To avoid losing of the airway
11. Reassess the airway patent and auscultated the lung for equal and clear	

breath sounds during ventilation.	
12. Record interventions that were implemented	

Endotracheal tube insertion and care

Outlines: -

- **Definition of endotracheal intubation**
- **Purpose of endotracheal intubation**
- **Contraindications of endotracheal intubation**
- **Complications of endotracheal intubation**
- **Equipment of endotracheal intubation**
- **Procedure of endotracheal intubation**
- **Procedure of endotracheal tube care**



Definition:

Endotracheal (ET) intubation involves the oral or nasal insertion of a flexible tube through the larynx into the trachea for the purpose of controlling the airway and mechanically ventilating the patient.

Purpose of endotracheal intubation:

- ✓ To keep the airway open to oxygen, medicine, or anesthesia.
- ✓ To support breathing in certain illnesses, such as pneumonia, emphysema, heart failure, collapsed lungs or severe trauma.
- ✓ To remove blockages from the airway.
- ✓ To allow the provider to get a better view of the upper airway.
- ✓ To protect the lungs in people who are unable to protect their airway and are at risk for breathing in fluid (aspiration). This includes people with certain types of strokes, overdoses, or massive bleeding from the esophagus or stomach.

Indications:

- ✓ emergencies such as cardiopulmonary arrest.
- ✓ Inability to maintain airway patency.
- ✓ Inability to protect the airway against aspiration.
- ✓ Failure to ventilate.
- ✓ Failure to oxygenate
- ✓ Epiglottitis.

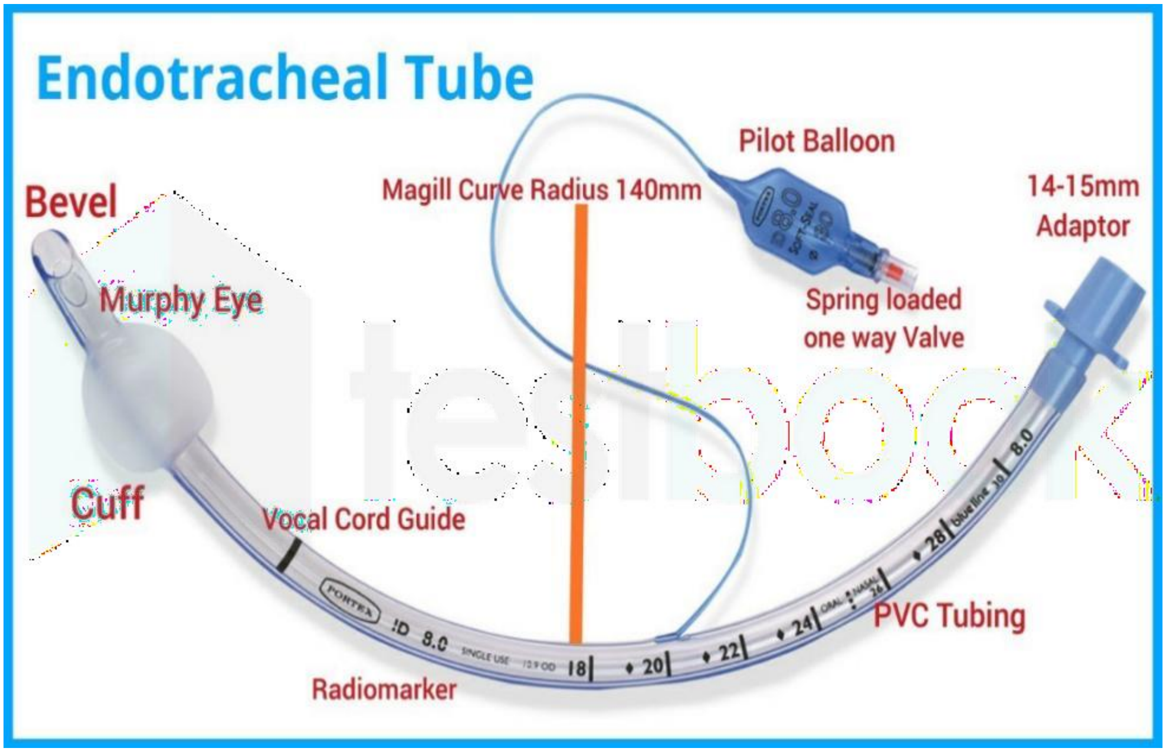
- ✓ Thoracic and abdominal interventions
- ✓ Head and neck surgery
- ✓ Muscle and central nervous system problems (Guillain-Barre, amyotrophic lateral sclerosis, myasthenia gravis)
- ✓ Respiratory deficiencies: patients with poor general condition, hypoxemic/hypercapnic

Contraindications: -

- ✓ severe airway trauma or complete upper airway obstruction
- ✓ tracheoesophageal fistula
- ✓ limitation of mandible movement
- ✓ The main contraindications to avoid placing an ETT with the nasotracheal approach include facial trauma, head trauma concerning basilar skull fracture, active epistaxis, expanding neck hematoma, oropharyngeal trauma, and apneic patients.

Complications of ETT intubation:

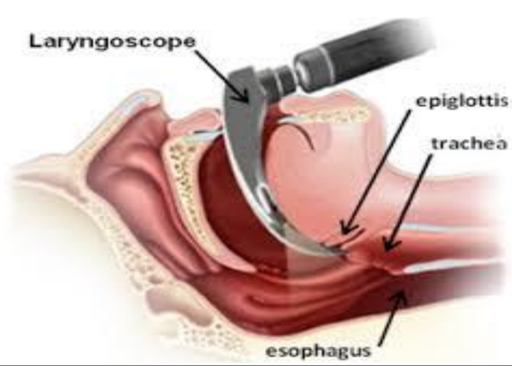
- Apnea caused by reflex breath holding or interruption of oxygen delivery; bronchospasm; aspiration of blood, secretions, or gastric contents; tooth
- damage or loss; and injury to the lips, mouth, pharynx, or vocal cords.
- Laryngeal edema and erosion and tracheal stenosis, erosion, and necrosis
- Nasotracheal intubation can result in nasal bleeding, laceration, sinusitis, otitis media, and VAP.


Endotracheal Tube
Pilot Balloon
Bevel
Magill Curve Radius 140mm
14-15mm
Adaptor
Murphy Eye
Spring loaded
one way Valve
Cuff
Vocal Cord Guide
PVC Tubing
Radiomarker

Equipment:

1. Personal protection equipment.
2. ETTs of appropriate sizes with intact cuff and connector.
3. Laryngoscope (to visualize the vocal cords) with fresh batteries and blades curved or straight.
4. Self-inflating manual resuscitation bag with mask connected to humidified oxygen source.
5. Oxygen source and connecting tubes.
6. 10cc Syringe for cuff inflation
7. Portable suction apparatus (ready with different catheter sizes for suction).
8. Oropharyngeal airway.
9. Endotracheal tube securing apparatus or adhesive tape or gauze for fixation.
10. Paralytic or sedative agent for intubation of combative patient (as valium).
11. Stethoscope.
12. Pulse oximetry to monitor oxygen saturation.
13. Stylet: flexible instrument inserted into the ETT to stiffen it to help direct insertion of tube to the glottic opening and avoid laryngeal trauma. 4. Local anesthesia (lidocaine, xylocaine spray) for

Procedure:

Steps	Rational
1) hand washing and wearing clean gloves and personal protective	To minimize spread of infection
2) Prepare equipment and check efficiency of the laryngoscope	To facilitate the intubation process 
3) Place clients in flat supine position with pillow under shoulders to hyperextend neck and help open airway	Proper positioning facilitates intubation 
4) Restrain client's wrists only if necessary.	
5) Give the sedations or the ordered medication according to the physician order	
6) Preoxygenate client for several minutes, using bag-valve mask	To create an oxygen reserve. 
7) Using thumb and index finger, apply cricoids pressure during tube insertion	This facilitates tracheal placement and

	protects against aspiration of gastric content 
8) Suction the airway if needed	Suction prevents the aspiration and makes air way clear
9) After ETT insertion attach bag-valve-mask, provide ventilation and look for chest to rise.	If chest doesn't rise, esophagus intubation is likely 
10) Auscultate lung fields for bilateral breath sounds.	This ensures that accidental right main stem intubation has not occurred.
11) Obtain chest x-ray	
12) Mark tube at level of client's front teeth and tape securely with twill or adhesive tape.	• Secure taping prevents tube displacement.
13 Place bite block or oral airway if ETT tube has been positioned orally	To avoid pitting on the ETT 
14-Attach O_2 source to ETT tube.	
15- In case of ventilation indication A-Confirm the ventilator moods and parameters with the physician	

B- Ensure that the humidifier has adequate fluid (sterile distilled water) and that the thermostat setting is adjusted.	Humidity is important because that decreases the risk of cilia injury.
16- Discard disposable equipment, remove protective gear, gloves, and perform hand hygiene.	
17- Documentation	

Endotracheal tube Care

Steps	Rationale
1. Wash hands and wear PPE	Minimize risk for infection
2. Prepare the needed equipment	To provide easy access and prevent interruption during procedure
3. Explain procedure to the patient as appropriate	Reduce anxiety
4. Note the proper tube insertion level in cm by making a landmark	The tube must not move back or forward during the procedure, thus patient may not receive the recommended oxygen therapy.
5. Auscultate air entry prior to procedure for comparison post procedure	Ensure that the tube is in proper place, if not reposition the tube properly.
6. Place a linen or towel on patient's chest	To keep the patient from being soiled
7. Turn on the suction device and adjust to appropriate setting	
8. Start by suctioning patient mouth and airway prior to procedure	To remove excess secretion
9. Prepare the tape to be used for resecuring the tube or a commercially prepared securing device	
10. With one nurse securely holding endotracheal tube, the second nurse carefully loosens securement device attached to the patient. If using adhesive tape method remove any excessive adhesive with adhesive tape remover and wipe off with wet face cloth.	
11. inspect the condition of face, lips and oropharynx for	

any break in integrity.	
12. Beginning on the site opposite to the placement of endotracheal tube, clean the patient mouth, gums and teeth. * Use soft bristle brush / oral swab and 10 cc syringe with oral cleaning solution. * Use the suction catheter as needed to remove excess secretions. * While cleaning check for any pressured area around the mouth.	
13. Note the endotracheal tube marking at the patient's lips, teeth and carefully move it to the opposite side of the mouth making sure it's depth remains the same.	
14. Clean the patient mouth, teeth and gum on this side. Suction if necessary.	
15. If tape is used, clean the area where the tape is attached and apply skin protectant as indicated.	
16. Resecure the tube by securing tape, confirm tube at requires cm marking prior procedure.	
17. Ensure that the tube is secure, it shouldn't move forward in the patient's mouth or backward down the throat.	
18. While ensuring, check for any pressure on the mucosa and around the occipital region on the head * If tape is used to secure the tube, place along tape underneath the patient's neck with the double portion resting from ear to ear. Secure one side of the tape over the lip and tube or from ear to nares. * Tear the remaining half lengthwise, secure the bottom half of the tape against the lip or across the top of the nose, wrap the top half of the around the tube twist to	

secure the endotracheal tube. Gently pull the other side of the tape that is around patient neck and secure to the opposite side of the patient's face and the endotracheal tube.	
19. Remove the linen from the patient and dispose properly all equipment used.	
20. Remove personnel protective equipments and wash hands.	
21. Document appropriate information: procedure time , any laceration found , secure level .	

Suctioning

Outlines: -

- **Definition of suction**
- **Indications of suction**
- **Types of suction**
- **Contraindications of suction**
- **Complications of suction**
- **Equipment of suction**
- **Procedure of suction**

Definition

Suctioning is a procedure used to remove mucus and secretions from the airway to maintain patency

and prevent respiratory complications.

Indications for Suctioning

- Noisy, gurgling breath sounds
- Visible or audible secretions
- Decreased oxygen saturation (SpO_2)
- Increased respiratory rate or effort
- Restlessness, cyanosis, or respiratory distress
- Artificial airway (endotracheal/tracheostomy tube)

Types of Suctioning

- **Oropharyngeal** – through the mouth (for conscious patients)
- **Nasopharyngeal** – through the nose (for unconscious or weak cough reflex)
- **Endotracheal** – through endotracheal tube (intubated patients)
- **Tracheostomy suctioning** – through tracheostomy tube

Equipment Required

- Suction machine and connecting tubing
- Sterile suction catheter (appropriate size)
- Sterile gloves (for sterile suctioning)
- Normal saline (for flushing)
- Oxygen delivery device
- Pulse oximeter
- PPE (mask, goggles/face shield, gown, gloves)
- Water-based lubricant (for nasopharyngeal suctioning)

Contraindications (Relative)

- Nasal trauma or surgery
- Coagulopathy or bleeding disorder

- Basilar skull fracture
- Severe bronchospasm

(Suctioning may still be performed with caution)

Complications

Complication	Rationale
Hypoxia / Hypoxemia	Suctioning removes air as well as secretions, leading to oxygen depletion.
Bradycardia *(vagal stimulation)*	Stimulating the vagus nerve during suctioning can trigger parasympathetic response → slow heart rate.
Tissue Trauma / Mucosal Injury	Repeated or forceful suctioning can damage the mucosa of the airway.
Bleeding (Epistaxis or Hemoptysis)	Trauma to fragile nasal or tracheal mucosa may cause bleeding.
Infection	Breach of aseptic technique or reuse of contaminated equipment can lead to respiratory infections.
Atelectasis	Excessive suction pressure or time can cause lung collapse by removing too much air from alveoli.
Increased Intracranial Pressure (ICP)	Coughing and straining during suctioning can increase ICP in neurologically compromised patients.
Bronchospasm or Laryngospasm	Airway irritation may trigger spasms, especially in sensitive individuals (e.g., asthmatics).
Anxiety or Discomfort	The procedure is uncomfortable and may cause distress, especially in conscious patients.
Vomiting and Aspiration	Gag reflex stimulation (especially during oropharyngeal suctioning) can lead to aspiration.

procedure

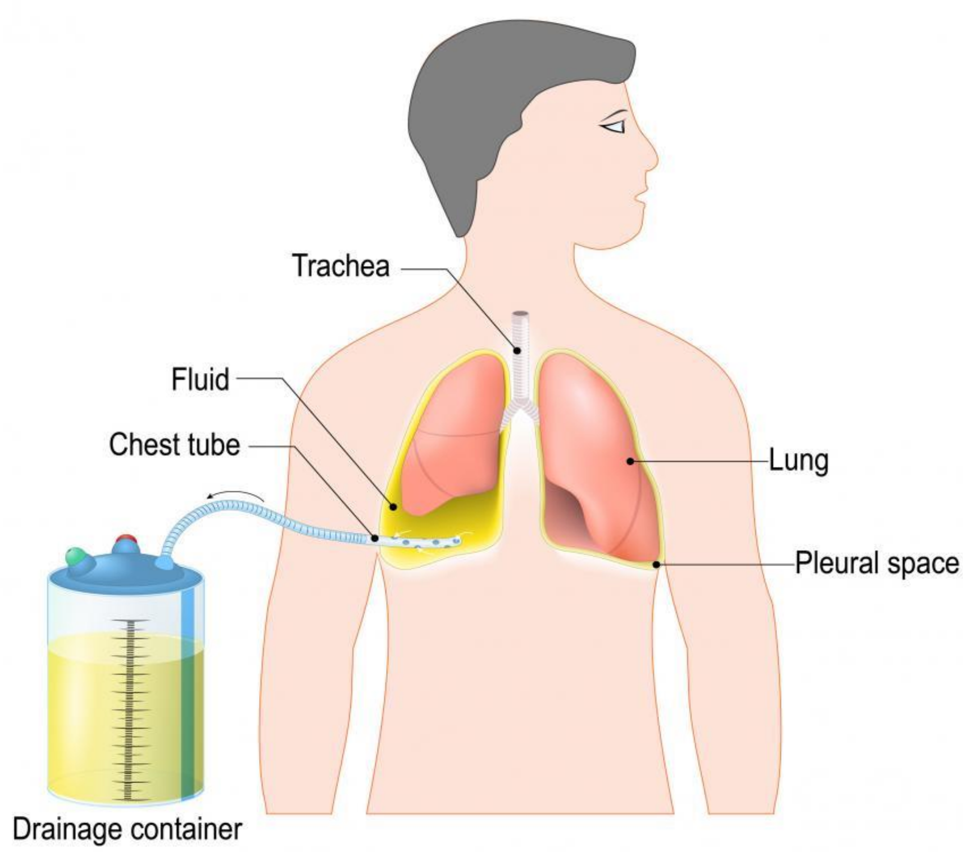
Step	Rationale
Verify physician's order (if required), assessing need for suctioning	• Ensures appropriate intervention
Explain procedure to patient	• Reduces anxiety and promotes cooperation
Perform hand hygiene and don PPE	• Prevents transmission of infection
Position patient in semi- or high Fowler's (unless contraindicated)	• Promotes lung expansion and access to airway
Check suction equipment; set to appropriate pressure: • Adults: 100–150 mmHg • Children: 100–120 mmHg • Infants: 80–100 mmHg	• Prevents trauma from excessive pressure
Attach pulse oximeter and assess baseline vitals	• Monitors for desaturation or distress
Provide 30–60 seconds of pre-oxygenation with 100% oxygen (if indicated, e.g., artificial airway)	• Prevents hypoxemia during suctioning
Open sterile suction kit and apply sterile gloves	• Maintains asepsis and reduces infection risk
Connect catheter to suction tubing without contaminating sterile end	• Ensures functionality without compromising sterility
Gently insert catheter without suction until resistance or desired length is reached	• Prevents mucosal trauma; allows accurate placement
Apply suction while withdrawing catheter using intermittent, rotating motion for ≤10–15 seconds	• Enhances secretion removal and reduces trauma
Flush catheter with sterile saline between passes	• Clears catheter and maintains function

Re-oxygenate the patient between suction passes for 30–60 seconds	• Allows oxygen saturation to recove
Repeat suctioning only if necessary (max 2–3 passes)	• Minimizes trauma and oxygen depletion
Dispose of catheter and gloves, turn off suction, and perform hand hygiene	• This step is critical to break the chain of infection, ensure safety, and complete the procedure according to standard nursing practice.
Document procedure and patient response ***1. Date & Time*** ***2. Type of Suctioning*** ***3. Indication*** ***4. Catheter Size and Type*** ***5. Suction Settings*** ***6. Number of Passes*** ***7. Pre-oxygenation*** *(if applicable)* ***8. Characteristics of Secretions*** ***9. Patient's Response*** ***10. SpO_2 Before and After*** ***11. Complications (if any)***	• 1.Legal and professional accountability • 2. Continuity of care • 3. Evaluation of patient response • 4. Early detection of complications • 5. Quality improvement and auditing

Chest Tube

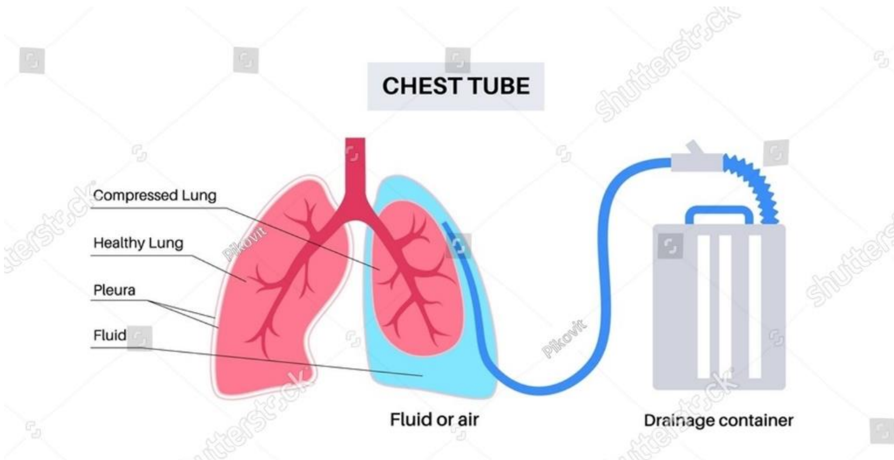
Outlines

- **Definition of chest tube**
- **Purpose**
- **Indications of chest tube insertion**
- **Sites of chest tube**
- **Types of chest tube bottle**
- **Contraindications of chest tube**
- **Equipment**
- **Procedure**



Definition:

A chest tube (chest drain) is a flexible plastic tube that is inserted through the Lateral/ or Mediastinal of the chest into the pleural space. It is used to remove excess air or fluid, or pus from the intrathoracic space and to help regain negative pressure.



Purposes:

- To Evacuate air and/or fluid from the chest cavity-
- To Evacuate fluid from around the heart (mediastinal) after cardiac surgery to prevent cardiac tamponade
- To Restore normal intrathoracic pressure (negative pressure)-

Indications:

- Pneumothorax:

There are two types of Pneumothorax:

- **Open Pneumothorax**: External trauma such as stab wound or bullet wound that penetrates chest wall may puncture lungs. Also called a sucking chest wound
- **Closed Pneumothorax**: Internal trauma i.e. Ribs fractures where rib punctures lung. No opening outside of the chest wall
- **Tension pneumothorax**: Occurs when air accumulates in pleural space more rapidly than it can be evacuated. Pressure builds up which not only causes lungs to collapse but can also shift mediastinum and severely impede venous return & cardiac output.
- Hemothorax
- Pleural effusions
- Chylothorax (a type of pleural effusion that results from lymphatic fluid (Chyle) accumulating in the pleural cavity).

- Chylothorax.
- Pyothorax or empyema (collection of purulent material in the lungs)
- Hydrothorax
- Pleurodesis (instillation of anesthetic medications or sclerosing agents)

Sites of chest tube insertion

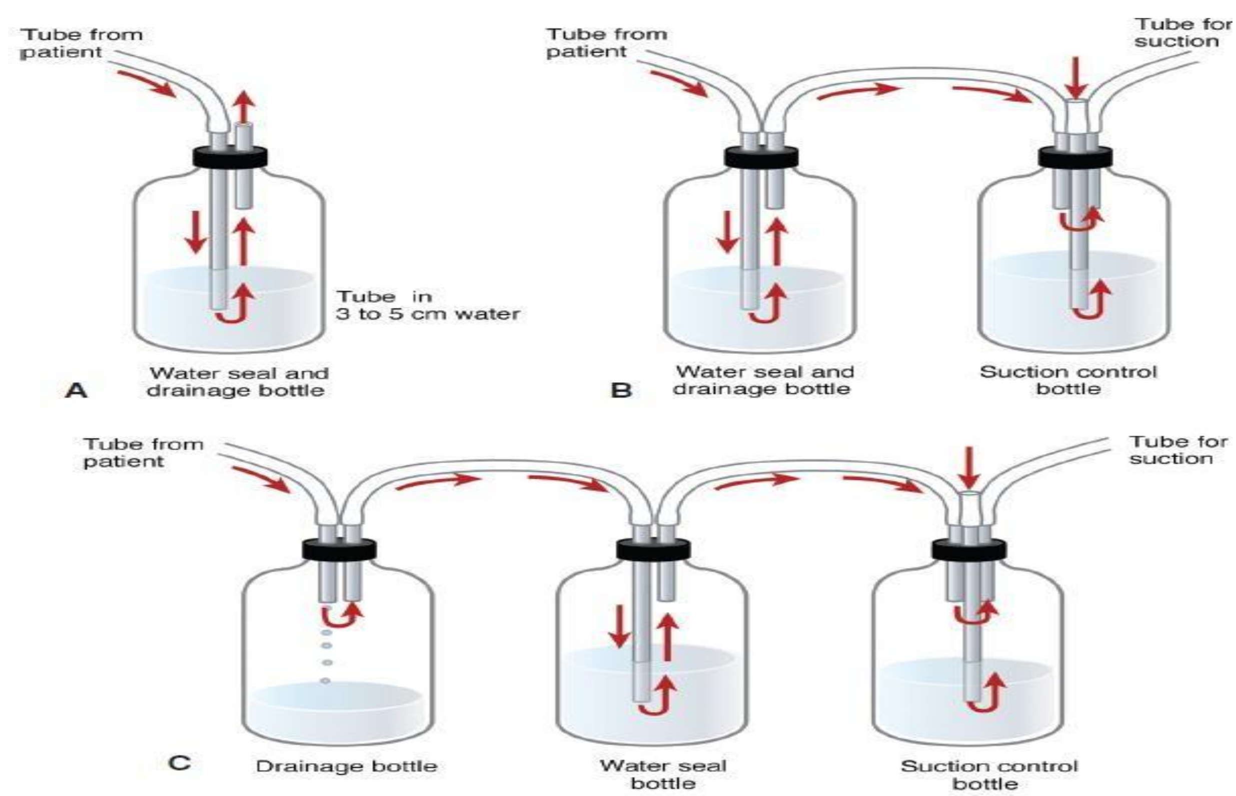
1. **For evacuation of air:** tube is placed near the apex of lung at **2nd intercostal space** are commonly used sites mid clavicular line
2. **To drain fluid:** tube is placed near the base of lung at **4th, 5th intercostal space** are commonly used sites, mid axillary line

Contraindications:

- **Coagulopathy**
- **Pulmonary bulla**: a small collection of air between the lung visceral pleura usually found in the upper lobe of the lung.
- **Pulmonary, pleural, or thoracic adhesions** -
- **Skin infection over the chest tube insertion site**

Types of chest tubes bottle drainage system:

1. The Single-bottled Water-Seal System
2. The Two-bottled Water-Seal System
3. The Three-bottled Water-Seal System



Source: D. J. Sugarbaker, R. Bueno, Y. L. Colson, M. T. Jaklitsch, M. J. Krasna, S. J. Mentzer, M. Williams, A. Adams: *Adult Chest Surgery*, 2nd Edition: www.accesssurgery.com

The Single-Bottle Water-Seal System

This system uses one bottle connected to the patient's chest tube. The drainage tube inside the bottle is submerged 2–3 cm below the water level, creating a water seal that acts as a one-way valve: air exits the chest but cannot re-enter. Drainage occurs by gravity. A second tube vents air through the water and out into the atmosphere. Bubbling may indicate a lung air leak or a system leak. The water level rises during inhalation and falls during exhalation. Continuous bubbling suggests ongoing air leakage. It's important to mark the water level before connecting to the patient to measure drainage accurately.

Two-bottle Water-Seal System:

This system includes a fluid collection bottle in addition to the water-seal bottle. Fluid drains into the first bottle, while the second bottle maintains the water seal. This separation allows for better fluid management. Like the single-bottle system, it relies on gravity. However, in cases of significant air leaks, suction may be needed—leading to the use of a third bottle.

Three-Bottle Water-Seal System:

This system consists of a collection bottle, a water-seal bottle, and a suction control bottle. The third bottle has three tubes: one from the water-seal bottle, one to the suction source, and one submerged in water (usually 20 cm) that opens to the atmosphere. The depth of this submerged tube regulates the suction pressure. Excess suction can harm the lungs. Bubbling in the suction bottle indicates the suction is functioning correctly.

Complications of Chest tubes

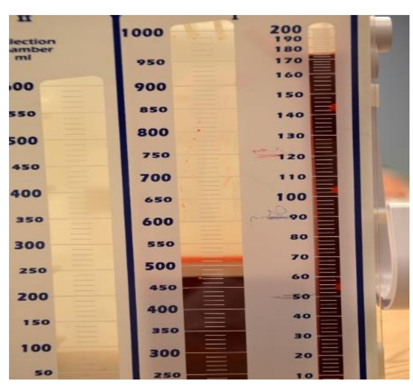
- Pain – chest wall/ neck / shoulder
- Failure to enter the pleural space
- Infection at insertion site or intra pleural
- Penetration / lacerations to lungs
- Hemorrhage
- Blocked drains
- Pleural sepsis
- **Subcutaneous emphysema**: occur when the lungs or the air passages are injured, air may enter the tissue planes and pass for some distance under the skin; the tissues give a crackling sensation when palpated. It is not a serious complication if the air is spontaneously absorbed or stopped, or if the leak is treated.

Equipment:

- Personnel protective equipment including sterile gloves.
- Sterile guaze.
- Adhesive tape .

Procedure for chest tube care:

	Steps	**Rational**
	Care:	
1-	Identify patient correctly	To do the correct procedure to right patient
2-	Wash hands and wear PPE	To prevent infection
3-	Explain procedure to patient	Reduce anxiety and gain patient help
4-	Remove dressing covering the chest tube insertion site	
5-	Using sterile technique, wrap petroleum gauze	

	around the chest tube insertion site.	
6-	Place a precut. Sterile drain dressing over the petroleum gauze.	
7-	Place a second sterile precut drain dressing over the first drain dressing with the opening facing in the opposite direction from the first one.	
8-	Place a large drainage dressing over the two precut drain dressing.	
9-	Secure the dressing in place with 2 inches silk tape, making sure to cover dressing completely.	
10-	Write date, time and initial on the dressing.	
11-	Remove personnel protective equipment and wash hands	To prevent infection
	Maintainance:	
12-	Assess patient every 1-2 hours for any changes.	Provides baseline and ongoing assessment of patient condition.
13-	Mark drainage level hourly or per shift on the drainage chamber.	Allows accurate monitoring of drainage amount and type, detects hypovolemia or bleeding. 
14-	Check for air leaks by observing bubbling and occluding tubing.	Identifies pneumothorax or leaks in the system or chest.
15-	Assess chest tube and system patency regularly; avoid routine stripping; gently milk if obstruction is visible.	Prevents obstruction and lung injury.
16-	Keep drainage tubing free of dependent loops.	Ensures unobstructed drainage and proper lung pressure.

17-	Monitor and maintain water-seal and suction fluid levels as prescribed.	Maintains correct suction and prevents complications.
18-	Observe tidaling/fluctuations in the water-seal chamber with respirations.	Indicates lung expansion and system patency.
19-	Inspect insertion site daily for crepitus, infection, and skin condition; change dressings when soiled.	Prevents infection and detects tube problems. 
20-	Monitor fluid volume in collection chamber; change system when full or damaged.	Maintains system integrity and patient safety.
21-	Keep system upright and below chest level during ambulation/transport; avoid clamping tube.	Ensures effective drainage and avoids tension pneumothorax.
22-	Assess pain and administer analgesia as prescribed.	Manages patient comfort.
23-	Obtain drainage specimen aseptically if needed.	Provides sample for laboratory analysis.

References

* **Accreditation Association for Hospitals and Health Systems. (2017).** Standard 07.01.21. *Healthcare Facilities Accreditation Program: Accreditation requirements for acute care hospitals.* Chicago, IL: Accreditation Association for Hospitals and Health Systems. (Level VII)

- **Ackley, B. J., & Ladwig, G. B. (2013):** Nursing diagnosis handbook: an evidence-based guide to planning care, Elsevier Health Sciences, (10thed.), 119-26.
- American Heart Association.
- **Bucher, L., Dirksen,S.R. , Heitkemper, M.M, and Lewis , S.L., (2014):**Medical-Sugical Nursing Assessment and Managment of Clinical Problems , (9th ed.) , USA:Mosby, 316-360.

* **Council Guidelines for Resuscitation (2015).** Section 8. Cardiac arrest in special circumstances, 2015;81:1400-1433.

* **Demir, M. (2023).**Cardioversion. In Updates on Cardiac Defibrillation, Cardioversion and AED Development. IntechOpen.

- **Ignatavicius,D.,&Workman,M.(2016).**Medical-surgicalnursing: patientcentered collaborative care (ed 8). St. Louis: Saunder
- **Layon AJ, Modell JH. Drowning: update 2009.** Anesthesiology . 2009;110:1390-1401
- **Lewis, S., Dirksen, S., Heitkemper, M., & Bucher, L. (2014).** Medicalsurgical nursing: assessment and management of clinical problems (ed 9). St. Louis: Mosby
- **Ralph, S.S and Taylor , C.M (2011) :** Nursing Diagnosis Pocket Guide, (1st ed.) , China : wolters kluwer health , (6,7,16,17,30,31).
- **The Joint Commission. (2019).** Standard RI.01.01.01. *Comprehensive*